



Probabilistic design and optimization of thermal protection system with variable thickness based on non-uniform aerodynamic heating

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ABSTRACT

Thermal Protection System (TPS) is a critical component of space vehicles to protect them from damage by extreme aerothermal heating conditions. The uncertainty of the aerothermal heating conditions, including approaching velocities, atmospheric density, pressure, etc., significantly affects the aerothermal heating imposed on the TPS, vastly changing the initial design conditions. In this study, we developed a new uncertainty quantitative (UQ) analysis method to comprehensively consider all the uncertainty of TPS materials, structures, and aerothermal heating conditions. To address the challenges of a massive amount of input and determined optimization parameters, we first induced the response surface method and Monte Carlo algorithm to produce a large number of analysis samples for Probabilistic Design. Then, the key affecting factors, not all the input design parameters, were chosen for optimization with sequential optimization and reliability assessment (SORA) strategy by a sensitivity analysis. With such improvements, the optimization for UQ analysis becomes practical. We used this new UQ method to analyze the effects of uncertainty of the aerothermal heating conditions on the thermal protection layer. A variable thickness design is proposed to adapt the non-uniform aerothermal heating conditions for the TPS. The results show that significant light-weighting can be achieved without the reliability penalty. The optimized structure also reduces internal temperature gradients and facilitates a more uniform temperature field distribution for the TPS. This study provides a new UQ engineering design approach for the system with many designed parameters.

1. Introduction

Thermal Protection System (TPS) is an essential part of space vehicles to protect them from extremely severe aerothermal heating in the atmosphere environment. The lightweight and thermal protection performance requirement will never be met simultaneously when space vehicles' speed increases [1,2].

The severe and uncertain aerothermal heating conditions make the thermal protection system design a significant challenge due to the complex aerodynamic enter/re-enter atmospheric parameters. Numerous uncertainties exist during the vehicle entry/re-enter process, such as minor deviations in flight trajectory, data measurement errors, material property dispersion, and aerodynamic environment [3]. Traditional design methods often assume that all factors affecting the safety of the aircraft occur simultaneously and with equal probability [4] without considering the likelihood of these factors occurring. Such

design leads to excessively high safety margins, resulting in oversized thermal protection system designs and impacting the aircraft's maneuverability. Therefore, addressing the uncertainties in various design parameters of the thermal protection system and environmental parameters during entry/re-enter poses a new challenge for the thermal protection system of space vehicles [5].

Probabilistic design (PD) method provides an uncertainty quantitative (UQ) solution for such an issue that is full of uncertainty, which uses a probabilistic approach to quantify stochastic uncertainty inputs and calculate the effect of input uncertainty on the output response. The probabilistic design can deal with uncertainty and risk [6,7,5] and reduce TPS's weight without losing the system's reliability [8,9]. For TPS, the uncertainties come from several aspects: 1) TPS materials properties, such as thermal conductivity, density, and thermal capacity; 2) geometry structural parameters, such as the shape and the thickness of the thermal protection layer; 3) atmospheric environment parameters, such as temperature, density, pressure, etc. The challenges of

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Nomenclature		TPS	Thermal Protection System
C	Specific heat capacity of the material, $\text{J (kg}\cdot\text{K)}^{-1}$	T_w	Outer surface temperature, K
$DENS$	Material density, kg m^{-3}	T_1	Atmospheric temperature, K
$Emis$	Material emissivity	u	velocity, m s^{-1}
H	Thickness, m	v_1	Flight velocity, m s^{-1}
h_d	the dissociation enthalpy, kJ kg^{-1}	θ_c	Angle of cone, $^\circ$
h_s	Stationary point enthalpy, kJ kg^{-1}	μ	Mean
h_w	Wall enthalpy, kJ kg^{-1}	ρ_1	Atmospheric density, kg m^{-3}
k	Thermal conductivity of the material, $\text{W (m}\cdot\text{K)}^{-1}$	σ	Standard deviation
Ma_1	Flight Mach number	τ	Time, s
P_1	Atmospheric pressure, Pa	Subscripts	
Q	Aerodynamic heating, W m^{-2}	0	Initial value
q_{cond}	Conductive heat flux, W m^{-2}	B	Inner surface
q_{conv}	Hot wall heat flux, W m^{-2}	EP	End point
q_{rad}	Radiative heat flux, W m^{-2}	IP	Intersection point
R_n	Inner radius of the head, m	MAX	Maximum value
SORA	Second Order Reliability Analysis	SP	Stagnation point
T_{crit}	Critical operating temperature of the material, K	T	Top surface

probabilistic design lie in the computation cost and efficiency since there are many possibilities when considering so many input uncertainties. Previous designs only thought the uncertainty of TPS materials' properties and geometry structures; few studies have investigated the impact of aerothermal uncertainties that arise after the vehicle has been put into service. Du et al. [10] proposed a thermal-elastic framework considering aerothermal uncertainties and ensured the reliable operation of the structure in uncertain thermal environments through design optimization. Wright et al. [11] reviewed the current state of aerothermal calculations and experimental analysis for missions entering Mars, emphasizing the importance of using Monte Carlo methods to identify and quantify physical model uncertainties. Studies also highlight the necessity of combining TPS material response with aerothermal predictions. Zhang et al. [3] used the K-L expansion method to represent the uncertainty of aerodynamic heating as a set of random variables. Results indicate that aerodynamic heating uncertainty exacerbates the randomness of TPS thermal response, significantly impacting TPS thermal reliability. Factors influencing the aircraft's thermal protection system's performance include material properties and structural dimensions and the flow velocity, temperature, environmental pressure, density, etc., during flight. Considering the impact of these aerodynamic environmental parameters during the design process poses a significant challenge for probabilistic design, requiring new approaches to deal with the huge computational resources required.

Without elaborately considering aerodynamic environmental parameters in the TPS probabilistic designs, the current PD of thermal protection systems are limited to one-dimensional, without combining non-uniform aerothermal effects imposed on the vehicle's surface. The aerothermal heating is not uniform along the vehicle's surface. Xu et al. [12] summarized that if the temperature field of steel structures exhibits noticeable non-uniformity, temperature-induced effects may occur, affecting structural performance and inducing safety hazards. Vennik et al. [13] conducted experiments in a pulse wind tunnel to reproduce non-uniform temperatures, proving that non-uniform heat flux significantly thickens the boundary layer near heated elements, and the effect of non-uniform wall temperature on adjacent flow differs from the effect of uniform wall temperature. Therefore, addressing the uncertainty of non-uniform aerothermal flux is crucial for enhancing system performance and mission success. The distribution characteristics of non-uniform heat flow, especially for surfaces such as the nose, wing leading edge, and areas with large heat-resistant tiles, can be optimized by considering the probability characteristics of different heat flux density distributions, adopting variable thickness design to balance

aerodynamic heating non-uniformity and structural lightweight [4,14,15]. Determining the appropriate thickness for different locations of large heat-resistant tiles [16] is extremely important for improving structural efficiency.

In addition, uncertainty analysis aims to establish the relationship between output parameters and uncertain input parameters, enabling multi-parameter optimization of input parameters. However, due to the vast computational requirements of probabilistic design, few studies have conducted a comprehensive probabilistic design that includes uncertainty optimization, especially considering the optimization for the confines of aerodynamic environment conditions. Zhu et al. [17] proposed a Six Sigma optimization method based on the Successive Response Surface Method, considering the uncertainty of material properties and geometric shapes. Hadi et al. [18] proposed a reliability-based multi-disciplinary design optimization method. Currently, most uncertainty-based optimizations for the Thermal Protection System (TPS) focus on dimension optimization under specified aerothermal loads. This implies that structural adjustments to the TPS do not alter the aerothermal loads on the surface [5,9,19]. Therefore, a comprehensive uncertainty-based optimization design that combines aerodynamic shape optimization with TPS structural optimization is more meaningful in practical engineering than any single optimization design.

In summary, the uncertainty and non-uniform aerothermal heating imposed on the performance of the thermal protection system require a new PD strategy and design concept for the TPS layers of space vehicles. This paper proposes a new method for PD design of TPS layer heating by non-uniform aerothermal heating conditions from two aspects, using surrogate models and Monte-Carlo simulations to solve the huge computational requirement and using parameter sensitivity analysis and sequential optimization and reliability assessment (SORA) strategy to deal with the UQ optimization. With such a PD method, a lightweight design method using variable thickness is proposed to adapt the non-uniform aerothermal heating conditions. With the optimization objective of minimizing total mass while ensuring reliability, uncertainty optimization is conducted to obtain structural dimensions of the thermal protection structure at different locations under non-uniform thermal loads.

2. Methodology

2.1. Aerothermal model

The instantaneous aerodynamic heating along the vehicle surface is an essential input for the PD of TPS [20]. The Fay-Riddell formula [21], the typical zero-angle-of-attack stagnation point of the vehicle, is used to calculate the aerodynamic heat from engineering aspects. It should be noted that the TPS we used as an example to show validation of probability design is the multi-layer containing silicon rubber-based thermal protection coating layer and other thermal insulating materials. Such TPS mainly works for the climb flight process of a hypersonic aircraft rather than the re-entry process. The consideration of angle of attack (AoA) might affect the accuracy of the aerothermal calculations for the re-enter process. However, for the climb flight process, the AoA has little effect on the aerothermal calculation. The use of Fay-Riddell formula has been proved to be capable of precisely calculating the aerothermal heating from engineering aspect. The general goal of this work is not to improve the accuracy of the aerothermal heating calculations, but how to characterize the uncertainty of imputed aerothermal heating and how to implete the uncertainty analysis to the TPS probabilistic design. The current multiple-input paratmers PD method can also be validated when we consider the effect of AoA.

$$q_s = 0.763 \text{Pr}^{-0.6} \left(\frac{\rho_w \mu_w}{\rho_s \mu_s} \right)^{0.1} \left(\rho_s \mu_s \frac{du}{dx} \right)^{0.5} \left(1 + (\text{Le}^{\text{Lea}} - 1) \frac{h_d}{h_s} \right) (h_s - h_w) \quad (1)$$

where Pr is the Prandtl number, Le is the Lewis number, and Lea is the weight of the Lewis number in the formula, q_s is the stagnation-point heat flux, ρ is density, μ is viscosity, h is enthalpy, du/dx is velocity gradient, h_d is the dissociation enthalpy, where subscripts w and s denote quantities at the wall and the stagnation point.

On the conical surface of a cone, a simplified formula for aerodynamic heating on the cone surface is obtained using the modified Newtonian pressure distribution and isentropic outflow conditions. This formula describes how aerodynamic heating varies with the position on the cone surface, providing crucial reference information for the design of thermal protection systems [22].

$$\frac{q_{wl}}{q_{ws}} = A(\theta_c) \frac{\left(\frac{x}{R_n} \right)^{0.5}}{\left[B(\theta_c) + \left(\frac{x}{R_n} \right)^3 \right]^{0.5}} \quad (2)$$

$$A(\theta_c) = \left(\sin^2 \theta_c - \frac{\sin^2 \theta_c + 1}{\gamma_1 Ma_1^2} \right)^{\frac{1}{2}} \cdot \sqrt{\frac{\pi}{2} - \theta_c} \cdot \frac{\sqrt{3}}{2} \quad (3)$$

$$B(\theta_c) = \frac{\frac{3}{16}}{\sin^2 \theta_c \left[\sin^2 \theta_c - \frac{\sin^2 \theta_c - 1}{\gamma_1 Ma_1^2} \right]} \left[\frac{D(\theta)}{\theta} \right]_{\theta=\frac{\pi}{2}-\theta_c} - \cot^3 \theta_c \quad (4)$$

$$D(\theta) = \left(1 - \frac{1}{\gamma Ma_1^2} \right) \left(\theta^2 - \frac{\theta \sin 4\theta}{2} + \frac{1 - \cos 4\theta}{8} \right) + \frac{4}{\gamma Ma_1^2} \left(\theta^2 - \theta \sin 2\theta + \frac{1 - \cos 2\theta}{2} \right) \quad (5)$$

where x' represents the distance along the cone surface from the fictitious cone apex, θ_c is the half cone angle, R_n is the inner radius of the head, γ is the gas adiabatic index, where θ is the angle between the flight direction and the radius vector from the center of the curvature of the nose, Ma_1 is the Mach number.

The calculation process of the aerodynamic heating calculation subroutine is shown in Fig. 1.

The accuracy of the aerodynamic heating engineering algorithm is validated using the blunt-nosed cone model from the NASATND-5450 report [23] and a substantial amount of accurate experimental data.

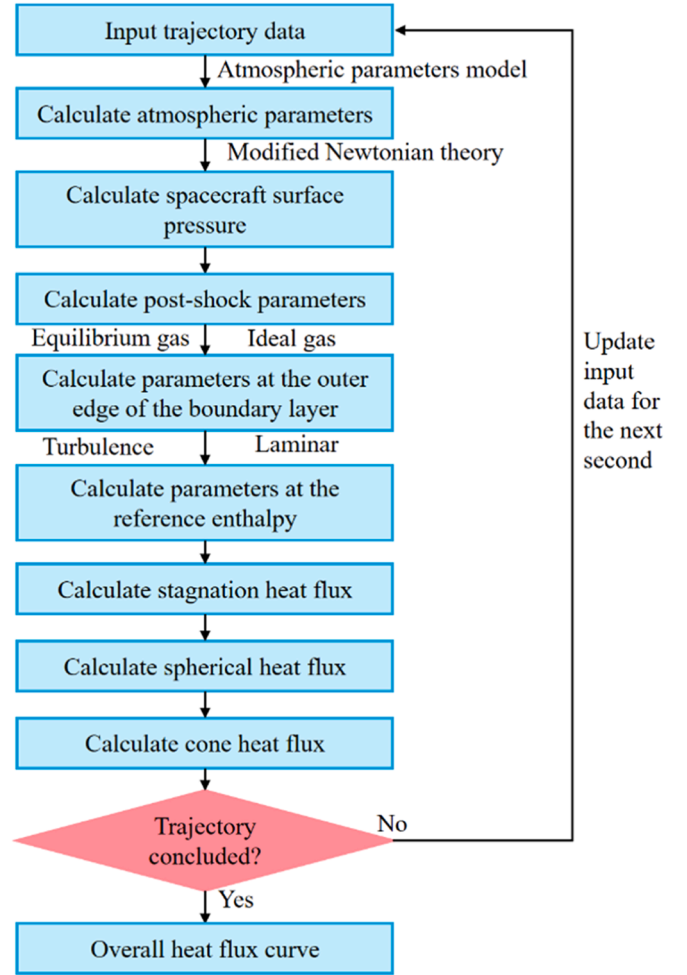


Fig. 1. Transient aerothermal calculation process flowchart.

The results in Table 1 indicate that the calculated stagnation heat flux error is within 4 %, demonstrating high accuracy from the engineering aspect. Furthermore, a comparison of the engineering algorithm-calculated simulated values at different positions on the cone with experimental data [6] shows the errors are within 9 %.

Considering the significant differences in thermal loads at different locations on the spacecraft, the lateral transfer of thermal loads is crucial in addition to the transfer of thermal loads to the internal structure. Therefore, selecting the spacecraft's most severely heated spherical end [15], a two-dimensional axisymmetric heat transfer model is established to effectively reflect the lateral transfer effects of thermal loads and ensure computational efficiency, as shown in Fig. 2. This model takes the stagnation point on the inner wall of the spacecraft's end as the origin. This study focuses on the climb flight process of the hypersonic aircraft's launch phase, in which the Mach number ranges from 0 to 7.14. Due to the short duration and low altitude, air dissociation phenomena have not been considered. Due to the short flight time and low altitude, the

Table 1

Verification of aerothermal engineering algorithms.

Region	Computed value	Reference value	Error
Stagnation aerodynamic heating (kW/m ²)	693.8	670	3.55%
	218.08	215.7	1.10%
	0.1302	0.122	6.7%
Blunt cone aerodynamic heating ratio	0.063287	0.0664	4.68%
	0.058	0.0636	8.86%
	0.0537	0.0581	7.54%

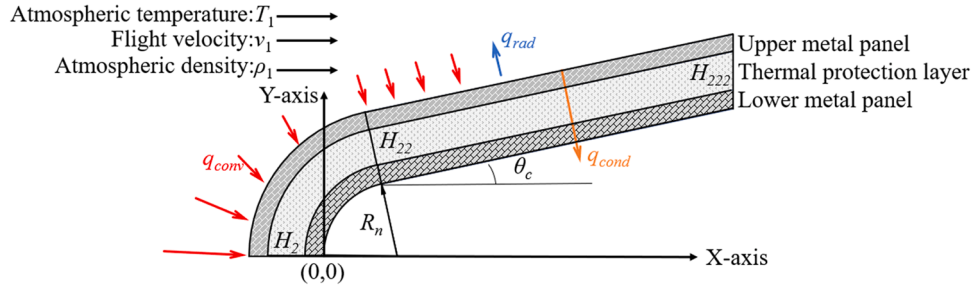


Fig. 2. Schematic diagram of thermal protection system structure.

enthalpy value of air dissociation is set as a fixed value of 137 kJ kg^{-1} . The aerothermal calculation method is not the focus of this paper, and the angle of attack is not considered as an input, assumed to be 0. The transient aerodynamic heat flux along the spacecraft's surface is calculated using Eqs. (1) and (2).

A three-layer thermal protection system is employed, with the outer panel using SiC composite material cover for resisting external aerodynamic heating. The intermediate insulating layer is made of a low thermal conductivity insulating material to prevent temperature transfer to the internal structure. The inner panel layer is made of quartz composite material to ensure the normal operation of instruments inside the cabin. The maximum temperature limit on the internal surface of the structure is set at 393 K. Material property parameters are shown in Table 2.

Performing aerodynamic heating numerical calculations for a fixed wall temperature (300 K) based on the flight trajectory shows that the maximum heat flux on the surface, i.e., the stagnation heat flux, reaches 80.96 kW m^{-2} . The heat flux gradually decreases linearly along the surface from the nose to the junction of the nose and cone, as shown in Fig. 3. However, there is a distinct inflection point at the transition between the spherical and conical sections, followed by a continued approximate linear decrease. Notably, the ratio of the heat flux at the cone tail over the stagnation heat flux is 0.127, indicating a highly uneven distribution of aerodynamic heating at different locations [15]. Therefore, considering the non-uniformity of heat flux is essential to improve structural efficiency.

2.2. Finite element modeling

Finite Element Analysis (FEA) [24] is used to solve the temperature distribution of the three-layer TPS.

The two-dimensional transient heat conduction equation can be expressed as:

$$\rho c \frac{\partial T}{\partial \tau} = \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) \quad (6)$$

The upper surface of the system is subjected to aerodynamic heating with the boundary condition:

$$k_x \frac{\partial T}{\partial x} n_x + k_y \frac{\partial T}{\partial y} n_y = q_{conv} - q_{rad} = q_{cond} \quad (7)$$

The lower surface of the system is treated as an adiabatic wall with

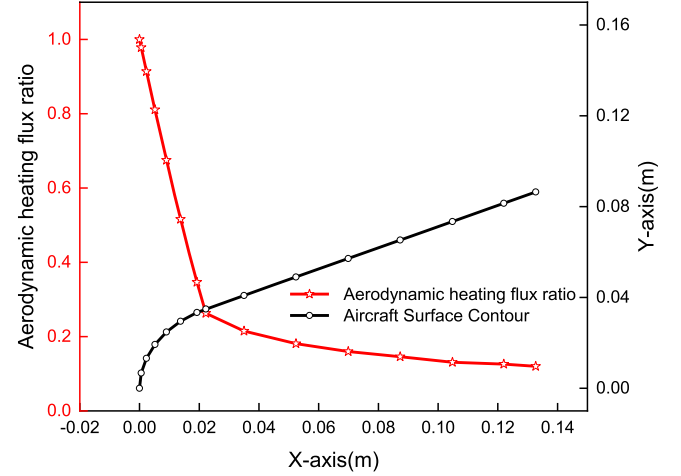


Fig. 3. Variation of non-uniform surface aerodynamic heating on the aircraft surface.

the boundary condition:

$$k_x \frac{\partial T}{\partial x} n_x + k_y \frac{\partial T}{\partial y} n_y = 0 \quad (8)$$

At the initial moment, the temperature distribution of the thermal protection layer is:

$$T(x, y, \tau) = T_0(x, y, 0) \quad (9)$$

where ρ is the material density; c is the material specific heat capacity; τ is time; k_x and k_y are the thermal conductivity of the material along the two principal directions of the object; n_x and n_y are the direction cosines of the outward normal to the boundary; T_0 is the initial temperature of the thermal protection layer, and T is the temperature, which is also a function of time and coordinates. By discretizing the partial differential equation using the finite element method, we obtain a system of ordinary differential equations:

$$CT' + KT = P \quad (10)$$

where C is the heat capacity matrix, K is the heat conduction matrix, both C and K are symmetric positive definite matrices; P is the load vector; T is the nodal temperature vector; T' is the vector of the derivative of nodal temperature to time. The solution of the above equations is accomplished using a direct integration method.

Based on the schematic diagram of the thermal protection system structure provided in Fig. 3 and the material parameter table (Table 2) for the insulation layer, we considered non-uniform heat flux loads along the outer surface in the X-direction. We established a finite element model with uniform thickness. Assuming no thermal resistance exists between structures, the outer surface is coated with an extremely thin layer with a radiative coefficient of 0.8. The initial temperature of

Table 2

Material properties.

Physical properties parameters	Upper metal panel	Thermal protection layer	Lower metal panel
Density (kg m^{-3})	2200	560	8100
Specific heat capacity (J ($\text{kg}\cdot\text{K}^{-1}$))	700	1510	500
Thermal conductivity (W ($\text{m}\cdot\text{K}^{-1}$))	16.7	0.3	10.7

the structure is set at 298 K. The maximum flight time is 181 seconds. We obtained the temperature distribution map through transient analysis at the final moment. We extracted the time-dependent temperature curves for three design points on both the upper and lower surfaces.

Fig. 4 shows the temperature contours and variation of the thermal protection system. Due to the gradually decreasing aerodynamic heat flux along the X-direction of the entire structure, the temperature gradient of the whole structure also decreases, resulting in 27.6 K mm^{-1} temperature gradients at the stagnation point, 15.5 K mm^{-1} at the junction, and 5.6 K mm^{-1} at the tail. The non-uniform aerothermal heating conditions lead to excessive temperature gradients, which might result in the generated thermal stress, affecting the material's performance and durability.

2.3. Probabilistic thickness design

Fig. 5 shows the proposed probability design process for the space vehicle's thermal protection system. It includes deterministic design, quantification of uncertain parameters, probability calculation, reliability assessment, and uncertainty optimization [25–27].

The PD starts with the Deterministic Design, establishing a finite element model to compute the thermal protection system's initial geometric dimensions and material properties. Subsequently, it is essential to identify the sources and distributions of uncertainty in quantifying uncertain parameter processes and incorporate them into the probabilistic analysis model. Uncertainties come from various sources,

- (1) Aerodynamic Heating Uncertainties [26]. During the enter/re-enter, the ionization and chemical reactions within the boundary layer introduce complex uncertainties. Additionally, variations in freestream conditions are significant contributors to aerothermal uncertainties. The dispersion of freestream parameters can reach up to 20% [26]. Therefore, aerothermal uncertainty is a crucial factor that needs careful consideration in the design of thermal protection systems [28].
- (2) Material Property Uncertainties [29]. This uncertainty primarily involves the dispersion of physical parameters such as thermal conductivity, specific heat capacity, density, etc. Typically, material uncertainties need to be determined by experimental measurements.
- (3) Geometric Shape Uncertainties. Such uncertainty comes from the thickness, geometric shape, and clearances introduced during the

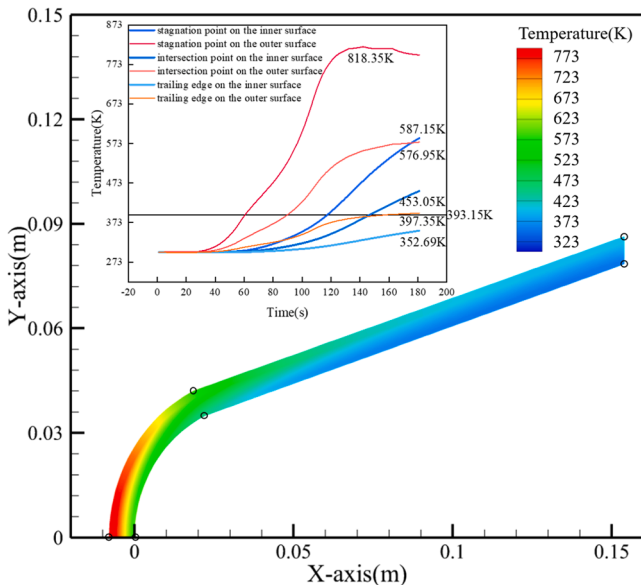


Fig. 4. Temperature contours and variation of the thermal protection system.

manufacturing and assembly processes, which can be obtained through measurements.

Thermal reliability analysis can be conducted using finite element analysis or surrogate models. In the presence of uncertainty factors, the thermal protection system's thermal reliability is defined as the probability that, within the specified duration of the flight mission and under the defined environmental conditions, the internal wall temperature of the spacecraft does not exceed a specified threshold.

$$R = 1 - P_F = 1 - \int_{T_{crit} \leq T_{MAX}} f(s) ds \quad (11)$$

where P_F represents the probability of failure, T_{crit} is the critical temperature the system can withstand, T_{MAX} is the maximum temperature of the system, and $f(s)$ is the probability density function of the random input variables.

In probabilistic design, establishing a reasonable reliability range is crucial. This range reflects the probability level at which the system meets the design requirements under uncertain confidence. However, a tradeoff between reliability and cost is necessary to avoid excessive reliability leading to unnecessary weight and cost.

Sensitivity analysis is used to assess the sensitivity of output parameters to changes in design variables [30], identifying uncertainty parameters with significant impacts on output parameters [31]. This guides the optimization direction of the thermal protection system, culminating in uncertainty optimization to obtain the optimal structural parameters for the thermal protection system's performance [32].

To address the challenges of traditional nested dual-layer optimization, the Sequential Optimization and Reliability Analysis (SORA) method has been introduced, significantly enhancing the convergence efficiency of optimization and providing a practical approach for the efficient solution of reliability optimization problems. The core idea of the SORA method is to decompose the nested problem of optimization-reliability analysis into two sub-problems: deterministic optimization and reliability analysis. These two sub-problems are then executed sequentially through iterative steps, ensuring the algorithm's convergence.

3. Results and discussion

3.1. Probability design of variable thickness under non-uniform heat flux

The distribution types and ranges of various uncertain parameters were set as shown in Table 3 [33–36]. Accurately estimating these uncertainty parameters is crucial for probability design and reliability assessment. These uncertainty parameters are input to the PD, generating the outcome of PD. The result of PD is the highest temperature on the inner wall of the spacecraft, T_{MAX} . With millions of input variables, a large number of random samples can be obtained from the statistical matrices representing each input variable. Each random sample will represent a set of uncertain parameter values. Subsequently, the corresponding T_{MAX} can be calculated using finite element analysis for each random sample. In this way, a series of T_{MAX} values corresponding to different combinations of uncertain parameters will be obtained. Once these T_{MAX} values are obtained, statistical analysis can be conducted, such as calculating the mean, variance, and other relevant statistics. These statistics can help us better understand the performance distribution and reliability of the system under uncertain conditions, providing crucial information for optimization design.

In thermal analysis, insufficient sample size may lead to false results in statistics. The challenge in uncertainty propagation analysis lies in the high dimensionality of uncertainty parameters and the high cost of analysis [3]. Directly conducting finite element uncertainty propagation analysis is time-consuming and inefficient. To address this, surrogate models can be employed to simplify computations. The Latin Hypercube

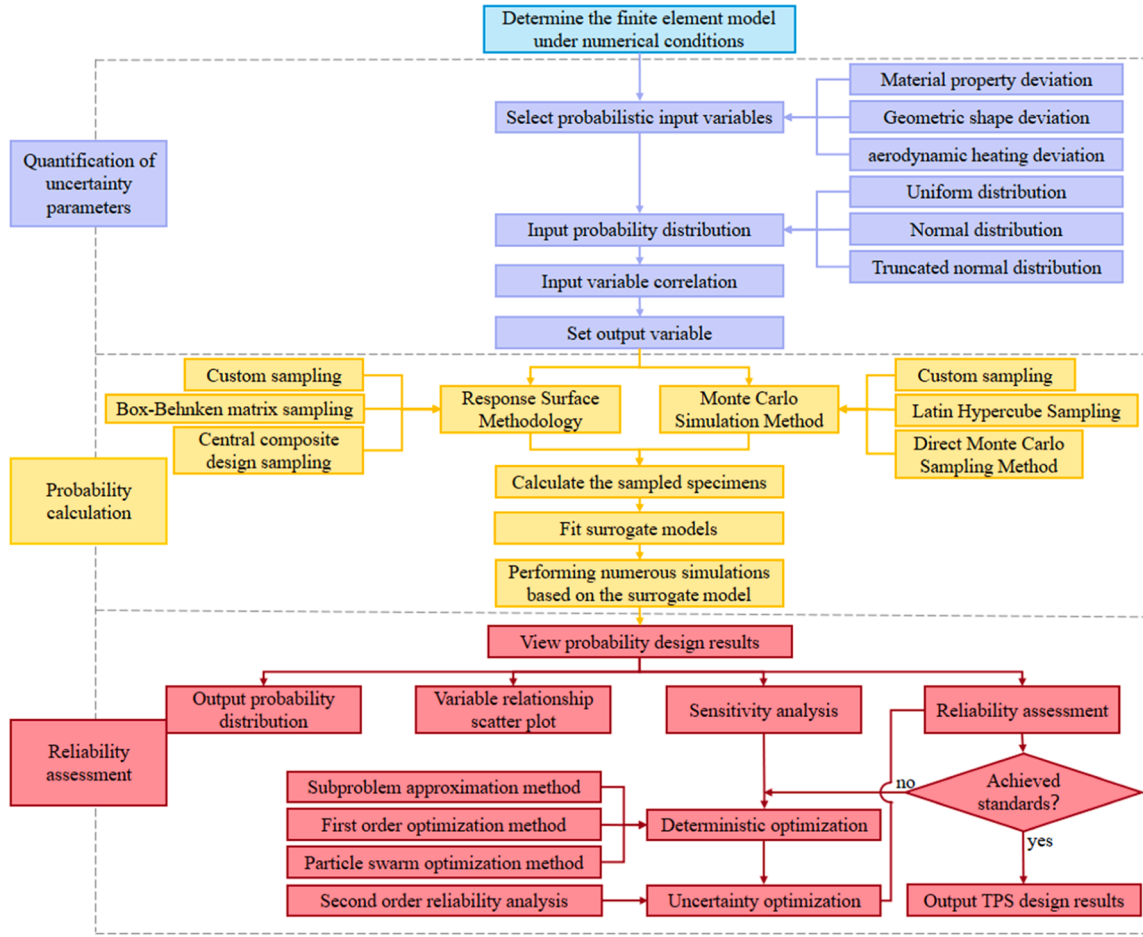


Fig. 5. Probability Design Process of the Thermal Protection System.

Table 3
Uncertainty input parameters.

Input variables	Variable symbol	Distribution type	Mean	Standard deviation	Lower limit	Upper limit
Thermal protection layer thickness at the stagnation point	H_2 (mm)	Truncated Gaussian distribution	11.5	0.345	10.35	12.65
Thermal protection layer thickness at the Intersection point	H_{22} (mm)		6.3	0.189	5.67	6.93
Thermal protection layer thickness at the end point	H_{222} (mm)		0.8	0.024	0.72	0.88
Inner radius of the head	R_n (mm)		38.5	1.155	34.65	42.35
Density of thermal protection material	$DENS_2$ (kg m ⁻³)		560	16.8	504	616
Specific heat capacity of thermal protection material	C_2 (J (kg·K) ⁻¹)	Uniform distribution	1510	45.3	1359	1661
Thermal conductivity of thermal protection material	k_2 (W (m·K) ⁻¹)		0.3	0.009	0.27	0.33
Flight speed coefficient	V_X		1	5%	0.8	1.2
Atmospheric density coefficient	M_X		1	5%	0.8	1.2
Atmospheric temperature coefficient	T_X		1	5%	0.8	1.2
Lewis number	Le	Uniform distribution			1.358	1.442
Lewis number weighting	Lea				0.5044	0.5356
Prandtl number	Pr				0.679	0.721
Initial temperature	T_0 (K)				293	303

Sampling [33] method was used to extract 150 samples based on Table 3. The highest temperatures on the inner wall of the structure are calculated as outputs for each sample. With these 150 input-output samples, a response surface surrogate model is then constructed, as shown in Fig. 6. Based on the surrogate model, 10,000 simulations were conducted, and the histogram and cumulative probability of the highest inner wall temperature were statistically obtained, as shown in Fig. 7. From the distribution of the inner wall temperature histogram, it can be observed that the temperature within the structure's inner wall approximately follows a normal distribution.

Sensitivity analysis was conducted to optimize multiple input parameters (more than 10) to identify the main matter uncertainty

parameters that most affect the internal wall temperature. By eliminating less influential parameters, the consideration of the entire set was avoided, preventing potential computational challenges. This ultimately serves to guide design and decision-making processes [31]. Fig. 8 shows the analysis results using the Spearman rank and Pearson linear correlation coefficient methods [37]. Based on these facts, the following selection can be made. The proportion of the impact on the bottom temperature by aerodynamic heating uncertainty, material property uncertainty, and geometric size dispersion is 3:1:1. The deviation in the spacecraft's flight speed has the most significant impact on the bottom temperature of the thermal protection system and is positively correlated. This implies that an increase in the deviation of flight speed leads

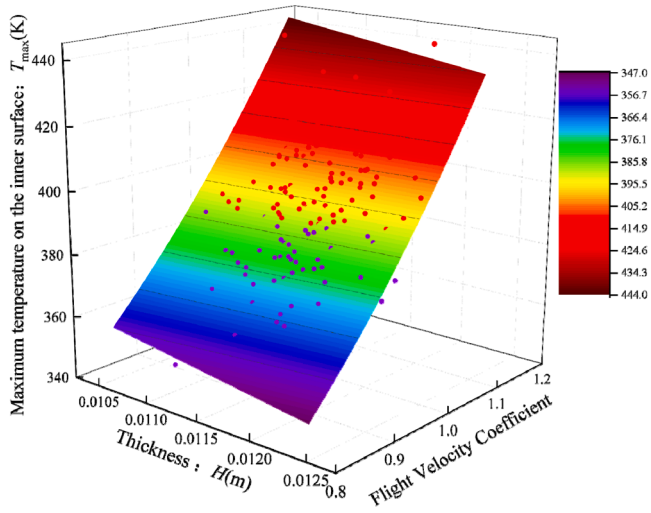


Fig. 6. Schematic diagram of the surrogate model.

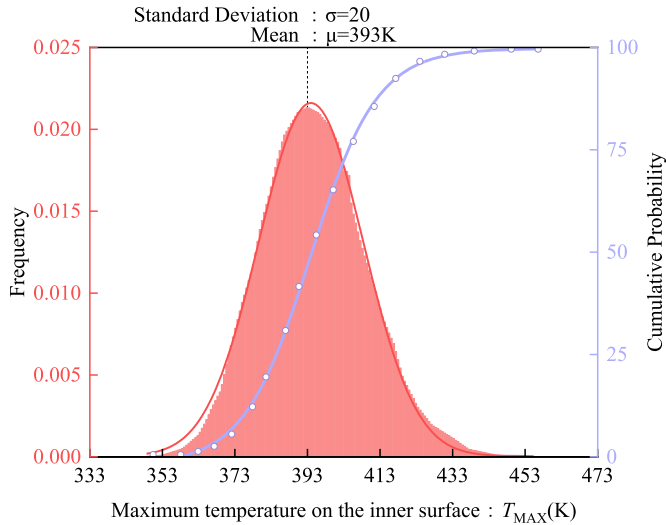


Fig. 7. Probability output of the temperature field.

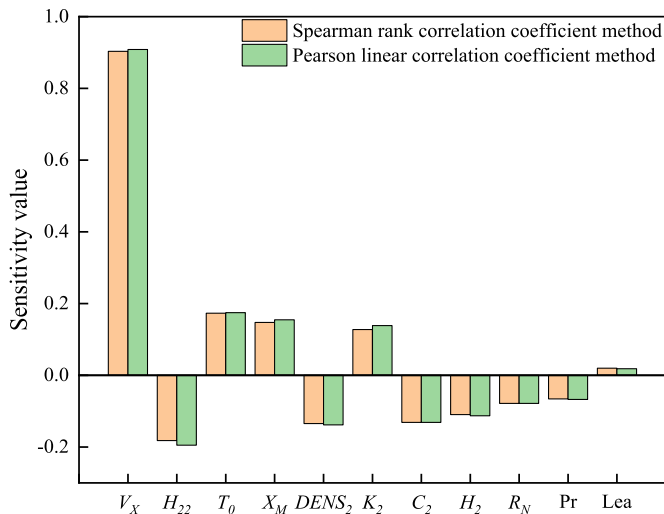


Fig. 8. Parameter sensitivity analysis.

to a rise in the bottom temperature, indicating poorer thermal protection performance. The thickness of the thermal protection layer at the stagnation point has the second-largest impact on the internal wall temperature of the thermal protection system and is negatively correlated. This suggests that an increase in the thickness of the thermal protection layer leads to a decrease in the internal wall temperature, indicating better thermal protection performance. The impact of other parameters is relatively small, in proportion to the following: initial temperature variation, atmospheric density deviation, thermal protection layer material density deviation, thermal conductivity deviation of the thermal protection layer material, specific heat capacity deviation of the thermal protection layer material, deviation in boundary thickness, deviation in inner wall surface spherical head radius, and cognitive deviation in the weight of the Prandtl and Lewis numbers. The deviations in the thermal protection layer thickness at the cone's tail, atmospheric temperature, and Lewis number are insensitive to the variation in internal wall temperature. Thus, they can be disregarded to reduce computational complexity.

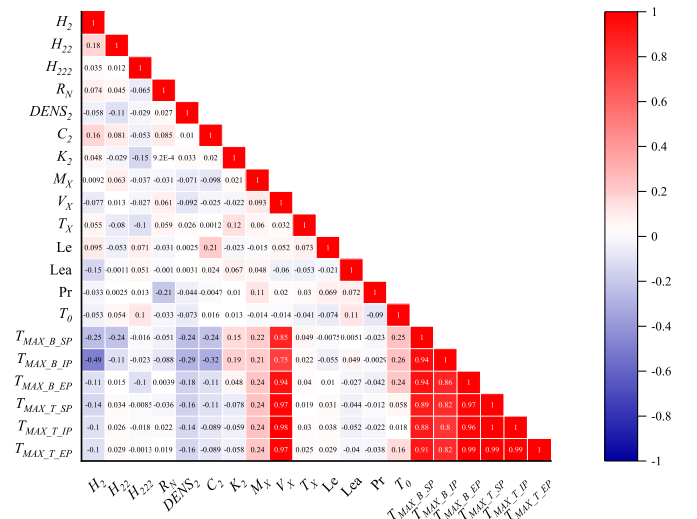
The parameter correlation coefficient between T_{MAX} and various inputs is shown in Fig. 9. With extensive Monte Carlo simulations, we can find that flight speed is the primary factor that significantly impacts the T_{MAX} for the variations in aerothermal parameters. It is essential to pay special attention to how these primary variations affect the system's performance in the design process. In addition, the input parameters also affect each other. These interacted uncertainty parameters propagation within the PD process. Therefore, the UQ analysis needs to consider the mutual relationships among various input parameters.

3.2. Reliability-based uncertainty optimization

We could process the PD optimization with the UQ analysis when we found the sensitivity of various input parameters. It should be noted that the general goal of the traditional PD method is only to find and manually adjust the sensitivity input parameters. Finding the optimal combination of the multiple inputs for TPS is a great challenge. For TPS design, lightweight is required with the constraint of safety. Therefore, the optimization goal function is defined as,

$$\begin{aligned} \min Mass &= f(H_2, H_{22}, H_{222}) \\ Prob\{T_{MAX}(H_2, H_{22}, H_{222}) \leq 393K \geq 0.9974(3\sigma) \\ 11 \leq H_2 &\leq 15mm \\ 5 \leq H_{22} &\leq 10mm \\ 1 \leq H_{222} &\leq 5mm \end{aligned} \quad (12)$$

We optimized the PD using the sequential optimization and

Fig. 9. Correlation coefficient between T_{MAX} and various input parameters.

reliability assessment (SORA) strategy. The optimal solution was obtained after five optimization iterations through the iterative adjustments of the SORA strategy. The optimal geometric parameters are obtained with $H_2 = 13.5$ mm, $H_{22} = 8.3$ mm, and $H_{222} = 3.6$ mm. Due to the changes in geometric thickness, the input aerodynamic heat along the surface also varied, exhibiting a complex nonlinear optimization issue. Ultimately, considering uncertainty, a design solution was found that not only meets the performance requirements but also minimizes the weight of the thermal protection system.

Fig. 10 shows the temperature contour of the optimal design solution. Compared to the original constant thickness design, the optimized structure exhibits a decrease in the protection side temperature, a reduction of TPS total mass, and a smaller temperature gradient within the thermal insulation layer along the flow direction. The optimization can also obtain a more uniform temperature field distribution. For the optimized structure, the temperature reliability of the innermost layer slightly exceeds the safety threshold of 0.9974. The result implies that for a confidence level of 95 %, the probability that the maximum temperature on the inner wall surface of the structure is below 393 K is slightly greater than 99.74 %, which can meet the requirements of the thermal protection system design from an engineering aspect.

To validate the correctness and efficiency of the uncertainty optimization results [34], a Monte Carlo method, in conjunction with finite element simulations, was conducted to generate 1000 random samples with mean values of 13.5 mm, 8.3 mm, and 3.6 mm, and a standard deviation of 3 %. With these samples, the SORA strategy was conducted with statistical analysis. It yielded the probability characteristics of the maximum temperature on the protection side wall surface of the spacecraft, as shown in Table 4. The results show that the reliability of the thermal protection system reaches 0.9978, satisfying the probability constraints on the thermal performance for each design requirement without introducing excessive redundancy. Compared to a single uncertainty analysis using the Monte Carlo method, the SORA method significantly reduces the overall optimization computational time.

With 1000 sets of Monte Carlo samples, we observed the mean and variance of the protection-side wall temperature throughout the entire spatiotemporal domain. The results, mainly focusing on the point of interest, are shown in Fig. 11. In contrast to calculations with Deterministic Design, the PD results are no longer represented by a single curve but in the form of mean and standard deviation, providing information about the uncertainty level of the output values. The blue curve is the mean of the protection-side wall temperature, with a

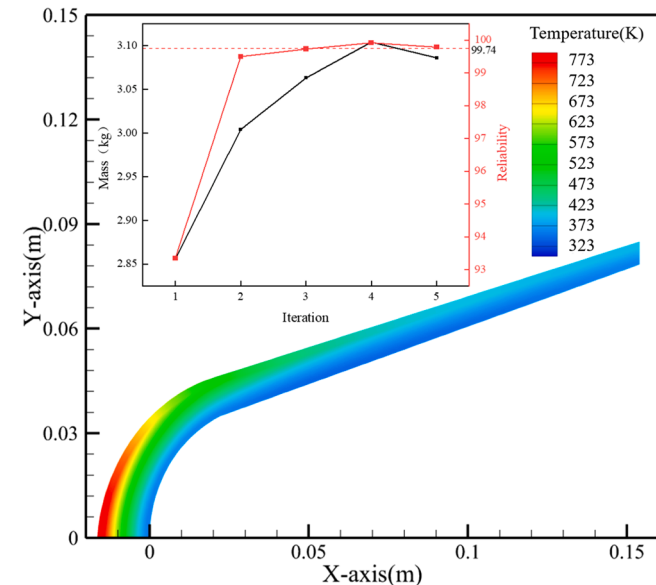


Fig. 10. Uncertainty optimization for variable thickness design.

Table 4
SORA optimization algorithm verification.

	Monte Carlo method	SORA method
Temperature reliability at the stagnation point	0.99787	0.99786
Temperature reliability at the intersection point	0.99788	0.99787
Temperature reliability at the end point	0.99816	0.99806
System mass (kg)	3.08624	3.08624
Time consumed (s)	70,000	52,500

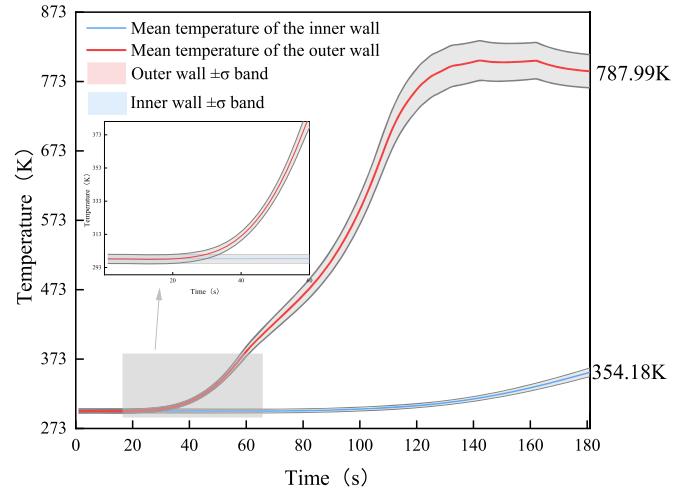


Fig. 11. Uncertainty temperature bands at inner and outer surfaces at stagnation.

maximum dispersion of approximately 7.68 %. The red curve is the temperature of the outer wall surface, with a maximum dispersion of about 4.64 %. The graph shows that the temperature uncertainty gradually increases over time, especially on the outer wall surface near the side of the heat flow input, where the temperature uncertainty level is the highest, confirming that greater uncertainty occurs where higher temperatures are exhibited [7]. Therefore, aerodynamic heating uncertainty significantly affects the uncertainties of inner and outer wall temperatures.

3.3. The advantages of variable thickness probability design

There are mainly two design methods for TPS with the basic idea of margin design [6,38], the Extreme Deviation (ED) method and the Root Sum Square (RSS) method. For the ED method, the required thickness margin of TPS is designed by the input parameters with the assumption to meet the worst-case scenario. For the RSS method, the uncertainties of both environmental and material parameters are separately calculated first to obtain the thickness margin for each uncertainty. The total thickness margin is then received by the square root of the sum of these two independent margins.

Based on the traditional ED Method and the aforementioned probabilistic design approach, we compared the following three design schemes: (1) Design #A is the conventional ED method, establishing a uniform thickness for non-uniform heat flux. In this method, if sensitivity is positively correlated, the upper limit of the parameters is chosen, but the lower limit is selected as the most unfavorable scenario if negatively correlated. (2) Design #B is also an ED Method for the variable thickness model with non-uniform heat flux. (3) Design #C is the PD method, the variable thickness model with non-uniform heat flux. The results are listed in Table 5.

Compared to the constant thickness (Design #A), the variable

Table 5
Comparison of results for different design scenarios.

Design scheme	Thickness (mm)			Mass (kg)
	The stagnation point	The intersection point	The end point	
A	16.8	16.8	16.8	3.8159
B	18	11.9	6.9	3.2770
C	13.5	8.3	3.6	3.0862

thickness (Design #B) based on the traditional margin design method reduces 16.44% of total mass. Variable thickness design can more precisely consider the non-uniform effects of heat loads. Compared to Design #B, the probability-based variable thickness design (Design #C) can further reduce the weight by 4.47% without sacrificing reliability. Probability-based design effectively reduces system weight while ensuring reliability by considering uncertainties and reducing redundant design in the structure, thereby decreasing weight and improving performance.

4. Conclusion

This work proposes a new PD design and optimization strategy for TPS in uncertain and non-uniform aerothermal heating conditions. A new design with various thicknesses is conducted to adapt to non-uniform heating conditions. The main conclusions are listed below.

- (1) The new PD method comprehensively considers all the uncertainty of TPS materials, structures, and aerothermal heating conditions. The response surface method and Monte Carlo algorithm replace the FEM to produce a large number of samples for PD. The key affecting factors, not all the input design parameters, were chosen for optimization with the SORA strategy by a sensitivity analysis. With such improvements, the optimization for PD becomes practical.
- (2) With the PD optimization, the probability that the maximum temperature on the inner wall surface of the structure is below 393 K is slightly greater than 99.74 % for a confidence level of 95

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijheatmasstransfer.2024.125386](https://doi.org/10.1016/j.ijheatmasstransfer.2024.125386).

Appendix Materials

S1: Deterministic Design

Traditional thermal protection system design models rely on specific numbers and deterministic values to describe and express the system. Material properties, component dimensions, and other parameters are input into the model with well-defined numerical values; this type of analysis is called deterministic analysis [2]. However, the results of deterministic analysis entirely depend on the accuracy of the numbers used and the assumptions made, as well as the applicability of these numbers under actual conditions.

Several key steps are involved in the deterministic design and analysis methods of the thermal protection system [39]:

Firstly, orbit design is carried out based on the mission requirements of the spacecraft, considering factors such as flight altitude, velocity, atmospheric conditions, navigation parameters, etc. Aerothermodynamic calculations are then performed to determine aerodynamic heating on the spacecraft's surface.

A design plan for the thermal protection system is formulated based on the distribution of external surface aerodynamic heating and temperature limits on the internal surface. This includes the structural form of various components and the materials used.

After determining the basic structure and materials of the thermal protection system, structural thermal analysis is conducted to determine the dimensions of the system, especially the thicknesses of various components. This process involves considering the temperature limits at different locations and ensuring that the maximum temperature on the internal surface of the thermal protection system meets the requirements.

$$h(x, y) = f(Q(t), MAT) \quad (13)$$

where $Q(t)$ represents aerodynamic heating, and MAT denotes material properties, $h(x, y)$ represents the thickness at the x, y position.

- (4) To account for the impact of uncertain factors on the thermal protection system's performance, it is common to introduce appropriate safety

%, which can meet the requirements of the thermal protection system design from an engineering aspect.

- (3) Compared to the traditional constant thickness, the variable thickness based on the conventional margin design method reduces 16.44 % of total mass. The probability-based variable thickness design can further reduce the weight by 4.47 % without sacrificing reliability. Probability-based design effectively reduces system weight while ensuring reliability by considering uncertainties and reducing redundant design

CRediT authorship contribution statement

Gui Lu: Conceptualization, Funding acquisition, Supervision, Writing – original draft. **Zhaoxu Shi:** Investigation, Writing – original draft. **Runlin Zhang:** Investigation, Methodology, Software. **Yuanyuan Li:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Kai Zhang:** Funding acquisition, Validation, Writing – review & editing.

Declaration of competing interest

The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Data availability

Data will be made available on request.

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margins on top of the nominal thickness. This process leads to the final dimensions and weight of the thermal protection layer.

(5) Finally, after the preliminary design meets the requirements, an assessment and validation process is conducted to ensure the performance and reliability of the thermal protection system.

The specific process is illustrated in Figure S1.

S2 Deterministic Optimization Design

Optimization methods aim to minimize or maximize the objective function under certain constraints. In previous research, various optimization methods have been proposed for structural design optimization [40,41].

The principles of lightweight design for the thermal protection structure can be summarized as follows: while satisfying the allowable temperature range of each material and constraints on inner and outer surface temperatures, the goal is to find the optimal combination of different thicknesses for each layer. This is done to achieve the overall lightest structure. Given that the insulating effect of the upper and lower panels is less significant, optimization primarily focuses on variations in the thickness of the insulating layer [24]. The following is a description of this optimization model:

$$\begin{aligned} \text{Min } Mass &= \sum_{i=1}^n m_i \\ \text{s.t. } \begin{cases} T_{MAX_B} \leq 393K \\ T_{MAX_T} \leq 1073K \end{cases} \end{aligned} \quad (14)$$

In the equation: i represents the i -th material, n is the total number of material types, and m_i is the mass of each material layer.

Intending to reduce the overall weight of the thermal protection structure, we extracted the maximum temperature values at each iteration step of the structural design. Subsequently, we employed subproblem approximation, first-order, and particle swarm optimization methods for optimization solutions. The results are shown in Figure S2, indicating that the optimal thickness decreases with the decrease in surface non-uniform aerodynamic heating [16].

S3: Surrogate Model

The response surface surrogate model assumes that a mathematical function can approximate the impact of random input variables on random output parameters. Therefore, the response surface surrogate model [42] is typically a quadratic polynomial.

$$Y = c_0 + \sum_{i=1}^{NRV} c_i X_i + \sum_{i=1}^{NRV} \sum_{j=1}^{NRV} c_{ij} X_i X_j \quad (15)$$

Here, c_0 is the coefficient of the constant term, c_i , $i = 1, \dots, NRV$ are the coefficients of the linear terms, and c_{ij} , $i = 1, \dots, NRV$ and $j = i, \dots, NRV$ are the coefficients of the quadratic terms. X represents the input variable and Y represents the output variable. Regression analysis is used to evaluate these coefficients, and the coefficients are typically evaluated to minimize the sum of squares of the differences between the true simulation results and the approximate function values.

Therefore, response surface analysis involves two steps [43]: (1). Performing simulation loops to compute the values of random output parameters corresponding to sample points in the random input variable space. (2). Performing regression analysis to derive the terms and coefficients of the approximate function.

It is important to note that for uncertainty propagation analysis, both the quantity and quality of samples are crucial. Therefore, before adopting a surrogate model, it is essential to carefully select appropriate sampling and construction methods to ensure that the surrogate model accurately captures the characteristics of the original high-precision model. The response surrogate model relationship diagram established in this article is shown in Figure S3.

To discuss the impact of the number of training samples and polynomial degree on the fitting results of the response surface surrogate model, four different response surfaces were constructed: linear polynomial response surfaces with training data $N = 100$ and $N = 150$, and nonlinear polynomial (quadratic) response surfaces with training data $N = 100$ and $N = 150$. The accuracy of each model fit is discussed below.

The accuracy of the model fit [44] is mainly measured by the adjusted R-squared (R^2_{Adj}) and the relative L_2 error. R^2_{Adj} is the coefficient of determination that considers the number of terms included in the regression equation. A value closer to 1 indicates a better model fit. The formula for calculating R^2_{Adj} is as follows:

$$R^2_{Adj} = 1 - \left[\left(\frac{SS_{residual}}{df_{residual}} \right) / \left(\frac{SS_{residual} + SS_{model}}{df_{residual} + df_{model}} \right) \right] \quad (16)$$

where $SS_{residual}$ and SS_{model} represent the sums of squares of the residuals and the model, respectively, and $df_{residual}$ and df_{model} represent the degrees of freedom of the residuals and the model, respectively. The relative L_2 error is a measure of the degree of difference between two sets of numbers. It can be expressed as follows:

$$L_2 = \| (y_{pred} - y_{true}) / y_{true} \|_2^2 \quad (17)$$

Where y_{pred} is the predicted or estimated value, y_{true} is the true value, and $\| \cdot \|_2$ represents the L_2 norm (Euclidean norm). The fitting accuracy results of the four response surfaces are shown in Table S1.

The table shows that the fitting goodness R^2_{Adj} and relative error L_2 of the nonlinear polynomial (quadratic) response surface with $N = 150$ are the best, indicating that better results can be obtained with more training data and higher-order polynomials. However, the linear polynomial with less data can produce good prediction results. In addition, the data size or order should be kept within an appropriate range to avoid issues like overfitting.

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